Online Testing of SMGF in ESOP Based Reversible Circuit

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***Abstract*—This work presents an online testing scheme for detecting Single Missing Gate Faults (*SMGF*) in Exclusive-Or Sum of Product (*ESOP*) based reversible circuits consisting of both the positive and negative control *MCT* (Multiple control Toffoli) gates. In this technique, initially we transform an input function to its equivalent testable design, where some redundant gates and an ancillary line are appended to the design. After that, the test vectors are applied to the input of the circuit for detecting all possible faults in the design. This scheme has been tested successfully over large benchmark circuits to verify correctness of the algorithm. Performance of the proposed design has been compared with the existing well known testing algorithms, and improvement over them has been observed.**

Keywords: -Exclusive-or sum of product (*ESOP*); Single Missing Gate Faults (*SMGF*); Missing gate faults (*MGFs*); Multiple control Toffoli (*MCT*)

I. INTRODUCTION

Heat dissipation is a major concern in present day's *VLSI* circuit. According to Launder's [1] findings, *KTlog2 joule* amount of heat is dissipated for loss of each bit of information (K = Boltzmann constant, T = absolute temperature). Bennett [2] postulated that a circuit constructed with reversible gates only can function without dissipating energy into the system.

Therefore, reversible logic computation may be the future solution for such type of logic design that will ensure minimum heat dissipation during logic computation [1, 3]. Applications of reversible logic is seen in various fields like quantum computing, cryptography, nanotechnology, *DNA* computing, optical computing and low-power *CMOS* design [4,5,6].

As the immense research is going on the synthesis and design of improved reversible logic, another issue is the testing of the reversible circuits. To address this issue, proper fault detection/diagnosis algorithms will have to be developed. The problems of fault modeling and testing of reversible logic circuits have been reported in several works [9, 10, 19, 24].

An adaptive tree based technique has been proposed in [11] for investigating the problem of "stuck-at" fault localization and detection in a reversible circuit. Universal testability of reversible logic circuits designed with *k*-CNOT gates under the stuck-at fault model has been proposed in [12].

A novel fault detection technique for reversible circuit is shown in [13], where some additional lines along with few redundant gates are appended in the design to convert it into an easily testable design. Though in this architecture some circuit overheads are generated but the testing process has become very simple.

Apart from these testing schemes, a new way of testing reversible circuit using Boolean difference method is presented in [14], where a Boolean generator is used to derive test sets which later are used to detect all type of *SMGFs* in reversible circuits.

Use of gate duplication strategy in testing of reversible circuit is presented in [16], where for each individual gate presents in a circuit upto 5 *k*-CNOT gates along with two redundant input lines are added to make the design easily testable one.

An efficient way of detecting faults in a special type of reversible circuit - exclusive-or sum of product (*ESOP*) is shown in [15]. In that work, Boolean expression based testing technique is developed that judiciously find outs the missing gate details along with the exact position of that as well.

All the works still so far stated generates sub-optimal solution space for testing but obtaining optimal solution is more complex. In recent year such a work is reported in [17], where optimal test sets (*OTS*) are generated for detection of multiple missing-gate faults in a reversible circuit.

Reversible circuit not only can be tested in offline mode but can be tested in online mode as well. This online mode has certain advantages over offline testing like as in online mode faults in the circuit can be tested without taking out the circuit from its operational mode. Since, last few years such works on online testing has been observed in reversible circuit testing. Here we are briefing some of them.

An online testing technique has been developed in [18], where online testability for an $(n\times n)$ reversible circuit is achieved using $2d$ garbage outputs and $(2d+1)$ extra input lines, where d was the depth of the circuits.

An online testing scheme for reversible circuit has been addressed in [19, 20], where three (4×4) reversible gates have been introduced for the testable design. Improved version of online testing scheme for reversible circuit has been introduced in [21, 22], where an improved version of checker circuit has been presented. A part from these works, an innovative way to trap faults in reversible circuits has been introduced in [23], where a dual rail coding approach has been used. A garbage free online testing scheme has been introduced in [24], where single missing gate faults in reversible circuit is detected without incurring additional overhead in the circuit in terms of garbage output.

In this work, we propose a new approach that detects *SMGFs* in *ESOP* based reversible circuits implemented with *MCT*

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gates. This online testing scheme has higher performance metrics compared to earlier develop online testing schemes. This scheme does not generate any additional garbage line in spite of appending one ancillary line in testable configuration. Improvement in quantum cost is also obtained in testable design.

The rest of the paper is organized as follows. A brief introduction on reversible family is presented in Section II. Details of proposed algorithm with example have been discussed in Section III. In Section IV, experimental results are given. Finally, Section V concludes the discussion.

II. BACKGROUND

A. *Reversible gates and circuits*

Definition1: A gate can be said a reversible one if the numbers of inputs are equal with number of outputs and perform one to one mapping between input to its output vectors.

Feynman [8], Toffoli [7] are the examples of reversible gate.

Definition2: When such reversible gates are arranged over a set of target and control lines then a circuit is formed and such circuits are called reversible circuits.

An example of a reversible gate is shown in Fig. 1(a), where the shown gate is 2-controlled Toffoli gate that acts over a single target line. In another figure (in Fig. 1(b)), a design of reversible circuit depicted that contains 4 reversible gate including reversible NOT gate.

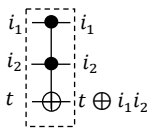

Fig. 1 (a): 2-controlled Toffoli gate

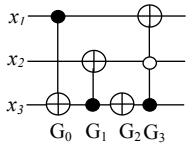

Fig. 1 (b): Example of reversible circuit

B. *ESOP based circuit*

The term ESOP stands for Exclusive sum of product. As like *Sum-Of-Product* (*SOP*), ESOP also demonstrates a functional form in Boolean expression where each product terms are separated by a XOR operator. For any reversible circuit, if logical expressions are obtained from their respective target lines then all the expressions will like as ESOP expressions. For example, the ESOP expression corresponding to the logical form $f = xy + yz$ is $f = xy \oplus yz \oplus xyz$.

Basically, such ESOP circuit are generated from cubelists which are collections of ESOP cubes. Such cube lists are represented in a matrix (M) form of size ($2^n \times n$), where each row represents a cube (C_i). Each cube C_i is mapped to a Toffoli gate T_i, where the value 0 corresponds to a negative control and bit value 1 represent a positive control. If the bit value is don't care (–) that means the control has no effect over the gate.

To represent a n-variable function in ESOP form requires a total of $(n + 1)$ circuit lines, where the fast n lines are control lines and the last $(n + 1)^{th}$ line is the target line from where the outputs are obtained.

In Fig. 1(c), a cubelist is given and its corresponding ESOP circuit is shown in Fig. 1(d).

x_1	x_2	x_3	x_4
–	1	1	–
–	–	–	0
1	–	–	–
1	1	–	1
–	1	1	0

Fig. 1(c): Cube list for $f(x_1, x_2, x_3, x_4) = x_2x_3 \oplus \bar{x}_4 \oplus x_1 \oplus x_1x_2x_4 \oplus x_2x_3\bar{x}_4$.

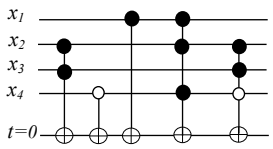

Fig. 1(d): Equivalent ESOP representation

C. *Fault Models*

A circuit can be said a faulty one if it does not produce the exact output that is desired to be obtained. This fault may be generated due to missing of gates or due to missing of control nodes in the circuit. Depending on this nature of faults [26], faults in reversible circuits can be categorized into three classes: Single Missing Gate Fault (*SMGF*), Partial Missing Gate Fault (*PMGF*) and Multiple Missing Gate Faults (*MMGF*). But the question comes that how such faults can be detected and it has been seen that the most usual way to test such faults is to test the circuit with test vectors which decides whether the circuit is faulty or not.

Such test vectors contain a certain Boolean patterns when they are applied over a certain circuit it checks the circuit correctness by judging the applied input pattern and the obtained output patterns.

Definition 3: If a single gate completely disappears from the circuit then this type of fault is known as Single Missing-Gate Fault.

See Fig. 2, where the SMGF is tested. Apply '1' to the respective control lines of the missing gates and fix the target line as well as all the through lines to either '0' or '1' to test the circuit.

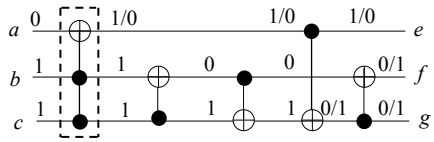

Fig. 2: *SMGF* in *ham3ndesign#1*

III. PROPOSED TECHNIQUE

Here, we discuss the online detection technique of *SMGFs* in *ESOP* based reversible circuits implemented with mixed polarity *MCT* gates. As the *ESOP* circuits can realize very large functions, in general such circuits contains a large number of *MCT* gates and due this reason, there are chances that these *ESOP* circuits may incur faults when it is in operational mode.

Such faults can be detected if some fault detection algorithms are applied over it after taking of the circuit from the operational mode. But without removing the circuit from operational mode fault can be detected if online testing schemes can be developed.

Here, we have developed such a testing algorithm that not only detects faults like *SMGF* in runtime but also detect it efficiently in *ESOP* circuits. The entire fault detection scheme

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is divided into three phases namely - *circuit augmentation phase*, *test vector generation phase* and *fault detection phase*. Now we are explaining all the phases sequentially in details with an example.

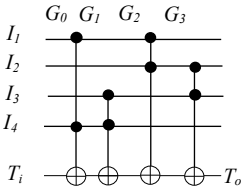

Fig. 3(a): Input *ESOP* circuit (C_{test})

I_1	I_2	I_3	I_4
1	–	–	1
–	–	1	1
1	1	–	–
–	1	1	–

Fig. 3(b): Equivalent Cube list corresponding to the input circuit (C_{test})

Phase 1: *Circuit augmentation:*

In this phase, an input *ESOP* circuit C_{test} (let it be the circuit as shown in Fig. 3(a)) is transformed to the testable circuit $C_{testable}$ after adding some redundant gates to the input design. This transformation is done in three steps.

Step 1: Append an ancillary line (*ANC*) just below the target line (T) in the input circuit C_{test}. See the modified circuit in Fig. 3(c), where the ancillary line is added to the input design of Fig. 3(a) but no connection is made with this ancillary line to the circuit.

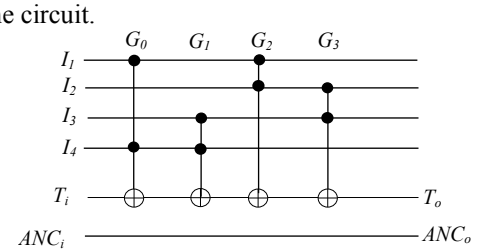

Fig. 3(c): Transformed circuit of Fig. 3(a) after executing step 1

Step 2: Now, we connect the newly placed ancillary line to the existing circuit C_{test} by appending some gates over the ancillary line. We have added the new gates in the circuit in such a way that for each *elementary gate* - G_i (those are present in the input circuit C_{test}), a similar gate - G_{ci} (*copy constituent gate*) is appended from the same control lines and is projected over the ancillary line *ANC*, where $i \in 0$ to n-1 and n is number of gates present in the circuit.

After execution of this step, the circuit of Fig. 3(c) is transformed to the circuit as shown in Fig. 3(d).

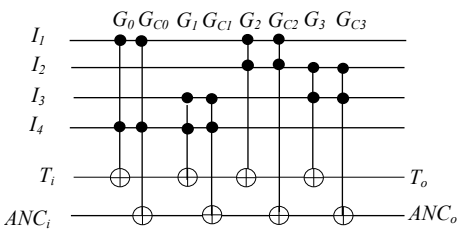

Fig. 3(d): Transformed circuit of Fig. 3(a) after executing step 2

Step 3: This is the final step of this phase and here we form the testable circuit ($C_{testable}$) which make the input circuit online testable. Here we add two additional *CNOT* gates G_{CN0} and G_{CN1} to the newly transformed circuit obtained in previous step in such a way that the first *CNOT* gate is placed before the first gate of the circuit and the second *CNOT* gate is placed after the last gate in that circuit. In our design, we term this pair of *CNOT* gates as "*target line supporting gate*".

Now, the circuit of Fig. 3(d) is transformed to the testable one ($C_{testable}$) in Fig. 3(e).

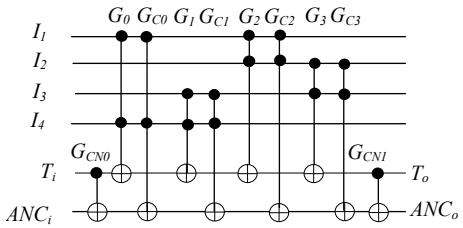

Fig. 3(e): Final testable circuit $C_{testable}$

Phase 2: *Test vector generation:*

Still so far we have generated the testable circuit and now here we are stating what type of test vectors can be applied over this testable circuit so that missing gate faults can be detected in online mode. This missing gate faults may happen due to missing of either *elementary gates* or *copy constituent gates* or due to *target line supporting gates.* But our developed scheme can handle missing of any type gates from the circuit.

The applied test vectors to detect the missing of either *elementary gates* or *copy constituent gates* are same and these test vectors can straightly be obtained from the *cube list* of input circuit.

If we assume that either G_0 or G_{C0} misses from the testable circuit of Fig. 3(e), then the cube <1- - 1> (as shown in Fig. 3(b)) corresponding to gate G_0 becomes the test vector for the circuit, where dash (-) sign represent the don't care situation i.e. we can place either 0 or 1 in any of the two vacant positions in the test vector. So, either of the four test vectors {<1 00 1>, <1 01 1>, <1 10 1>, <1 11 1>} can be applied at the input $<I_1,I_2,I_3,I_4>$ of the testable circuit to detect the fault.

If the *target line supporting gates* go missing then the following two test vectors have to be applied at the input section $<I_1,I_2,I_3,I_4,T>$ of the circuit.

Apply test vector <- - - - 1> (where 0, 1∈ -) if the first one (G_{CN0}) goes missing and apply test vector <00001> if the last one (G_{CN1}) misses.

Phase 3: *Fault detection phase*:

Here, we check whether the input value (ANC_i) of ancillary line (*ANC*) is same with the output value ANC_o or not for a specific test vector. If both the values i.e. ANC_o and ANC_i for that test pattern are equal, then there are no faults in the circuit else definitely, the circuit has a fault.

It can be observed that the *ESOP* expression ($T_i \oplus I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3$) generated from output line (T_o) of Fig. 3(e) is same as the *ESOP* expression from output line (T_o) of the circuit as shown in Fig 3(a). This reveals that the testable design is not affecting the original *ESOP* circuit function output.

If there does not exists any *SMGF* in the testable circuit ($C_{testable}$), then *ESOP* expression from the output line ANC_o would be same as input value in the port ANC_i i.e. $ANC_o = (ANC_i \oplus T_i) \oplus (I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus (I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus T_i = ANC_i$.

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Now we are considering the situation when there exist faults in the circuit.

Let us assume that the gate G_0 is missing from the testable design ($C_{testable}$) as shown in Fig. 3(e) and this missing gate fault information can be retrieved if we collect the value from port ANC_o and analyze it. If the value at port ANC_o is reverse to value of port ANC_i, then definitely there is an error in the testable design.

Here, we are stating the various cases when particular typical gate goes missing.

Case 1: When *elementary gates* in the testable design go missing:

Let a certain condition when gate G_0 in Fig. 3(e) goes missing from the circuit $C_{testable}$ by making it a faulty one. The expression from output port (ANC_o) of anicallary line ANC in circuit $C_{testable}$ become $(ANC_i \oplus T_i) \oplus (I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus (I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus T_i = ANC_i \oplus I_1I_4$. Now apply test vector $<I_1,I_2,I_3,I_4> = <1- -1>$ (where 0, 1∈ -) over circuit $C_{testable}$.

So, $ANC_o = ANC_i \oplus 1.1 = \overline{ANC_i}$

Previously, we have seen that in non faulty circuit the output port value (ANC_o) was same as input port value (ANC_i) but when there was a fault in the circuit ($C_{testable}$) the value has toggled. This very fact is the indicator of faults in an input test circuit.

Similarly, information regarding missing of remaining gates in Fig. 3(e) can be obtained using the previously stated way.

For missing of gate G_1:

Output line expression $ANC_o = ANC_i \oplus I_3I_4$. Applied test vector over circuit $C_{testable}$is <- - 11> (where 0, 1∈ -), which changes ANC_i to $\overline{ANC_i}$.

For missing of gate G_2:

Output line expression $ANC_o = ANC_i \oplus I_1I_2$. Applied test vector over circuit $C_{testable}$ is <11 - -> (where 0, 1∈ -), which changes ANC_i to $\overline{ANC_i}$.

For missing of gate G_3:

Output line expression $ANC_o = ANC_i \oplus I_2I_3$. Applied test vector over circuit $C_{testable}$ is <- 11 -> (where 0, 1∈ -), which changes ANC_i to $\overline{ANC_i}$.

Still so far we have discussed how the missing of *elementary gates* from testable *ESOP* circuit can be detected while the circuit is in operational mode. But one very important concern arises when the redundant gates (*copy constituent gates* and *target line supporting gates*) that we have appended in the input test circuit go missing. To address this aspect, we have come with a strategy as like previous one where single missing gate faults in *elementary gates* are detected.

Case 2: When *copy constituent gates* in the testable design go missing:

The proposed algorithm that we have introduced is self-sufficient to detect such type missing gate faults. To justify our claim here we have shown that particular scenario when a *copy constituent gate* (let G_{C0}) goes missing from testable design ($C_{testable}$) in Fig. 3(e).

Due to missing gate G_{C0}, the output expression ANC_o become $(ANC_i \oplus T_i) \oplus (I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus (I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus T_i = ANC_i \oplus I_1I_4$ and after applying test vector <1- -1> (where 0, 1∈ -) over circuit $C_{testable}$, we find that port value in line ANC_o has toggled i.e. $ANC_o = ANC_i \oplus I_1I_4 = \overline{ANC_i}$.This very fact confirms that there is a missing gate fault in circuit $C_{testable}$.

Similarly, missing of other *copy constituent gates* like G_{C1}, G_{C2}, $G_{C3}$from circuit $C_{testable}$ in Fig. 3(e) can be tracked. Here, a very interesting fact can be observed that the applied test vectors for missing of *copy constituent gates* are same as test vectors for missing *elementary gates.*

Case 3: When *target line supporting gates* in the testable design go missing:

If gate G_{CN0} goes missing then the support line expression become $ANC_o = (ANC_i \oplus I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) \oplus (T_i \oplus I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3) = ANC_i \oplus T_i$. If we set target line (T) input port value $T_i = 1$, then for the test vector $<I_1,I_2,I_3,I_4,T> = <- - - - 1>$ (where 0, 1∈ -) over circuit $C_{testable}$, $ANC_o = \overline{ANC_i}$, which reveals the missing of information of gate G_{CN0} from testable design.

If gate G_{CN1} goes missing then we obtain $ANC_o = (ANC_i \oplus T_i) \oplus (I_1I_4 \oplus I_3I_4 \oplus I_1I_2 \oplus I_2I_3)$. After apply the test vector $<I_1,I_2,I_3,I_4,T> = <00001>$ through the circuit, we find the missing gate information as we see $ANC_o = \overline{ANC_i}$.

Another example is given here where the entire fault-detection scheme has been explained.

Example 1: Now, consider the *ESOP* circuit and its corresponding *cube list* as depicted in Fig. 4(a) and Fig 4(b) respectively. To make the circuit online testable, the design of the original circuit is modified according to the different phases of the proposed algorithm and the testable design corresponding to the input circuit (in Fig. 4(a)) is shown in Fig. 4(c).

The output expression T_o from the circuit in Fig. 4(a) is, $T_o = T_i \oplus I_1\overline{I_4} \oplus I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}$

The output expression T_o from the circuit in Fig. 4(c) is $T_o = T_i \oplus I_1\overline{I_4} \oplus I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}$

Hence it can be seen from these two expressions that the circuit functionality retain same in the testable design (in Fig. 4(c)) after addition of redundant gates into the original one (in Fig. 4(a)).

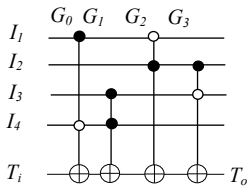

Fig. 4(a): Input ESOP circuit (C_{test})

I_1	I_2	I_3	I_4
1	–	–	0
–	–	1	1
0	1	–	–
–	1	0	–

Fig. 4(b): Cube list for the Fig. 4(a)

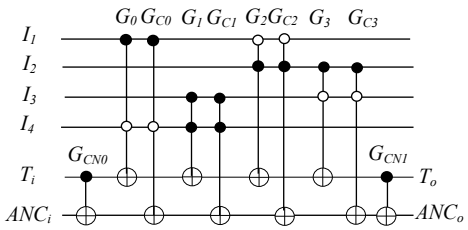

Fig. 4(c): Equivalent testable representation of Fig. 4(a) ($C_{testable}$)

For the fault free circuit of Fig. 4(c), the value at port ANC_o is given by, $ANC_o = (ANC_i \oplus T_i) \oplus (I_1\overline{I_4} \oplus I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}) \oplus (I_1\overline{I_4} \oplus I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}) \oplus T_i = ANC_i$ and it also can be seen that the input port value (ANC_i) at line ANC in the testable design is same as the output value (ANC_o) in that ancillary line.

Now, let us assume that the elementary gate G_0 is missing from the testable design of Fig. 4(c). Due to the missing of gate G_0, the output expression ANC_o would be as given below: $ANC_o = (ANC_i \oplus T_i) \oplus (I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}) \oplus (I_1\overline{I_4} \oplus I_3I_4 \oplus \overline{I_1}I_2 \oplus I_2\overline{I_3}) \oplus T_i = ANC_i \oplus I_1\overline{I_4}$.

While applying test vector $<I_1,I_2,I_3,I_4>$ = <1- - 0> over the output expression ANC_o, we find that $ANC_o = \overline{ANC_i}$ which confirms the presence of fault in the input test circuit.

For missing of other *elementary gates* like G_1, G_2 and G_3, faults in the circuit can perfectly be detected by test vectors < - -11>, <01 - - > and < -10 ->.

If *copy constituent gates* misses from the testable realization, then it also can be detected like the way that already have said and the applied test vector would be same as the test vector applied to detect missing of corresponding elementary gates.

Similarly for missing of all individual gate faults (irrespective of nature of gates, it might be *target line supporting gates* or *copy constituent gates* or *elementary gates*) can be detected using appropriate input test vectors.

IV. Experimental Result

The algorithm that we already have introduced in previous section has been successfully tested over benchmark [28] circuits to check functional correctness. The obtained experimental results have been presented in Table 1, where comparisons have been made with experimental data that are reported in [19], [20], [23] and [24]. The two metrics quantum cost (QC) and garbage outputs (GO) is considered in comparison analysis. It can be seen from the Table 2 that our approach achieves 34.05%, 44.03% and 3% improvements by taking quantum cost as base metric and an improvement of 99.84%, 98.01% is achieved when the metric garbage output is considered. The circuit overhead of testable design and the non-testable design is also computed in Table 3. The proposed algorithm also performs better than the technique as stated in [27] and the supporting results are tabulated in Table 4.

V. Conclusion

This work has presented an online fault detection technique under single missing-gate fault (*SMGF*) model for *ESOP* based reversible circuit. The developed scheme not only can detect faults in run time mode but also is applicable to any size of input circuit. We have applied our testing schemes over several benchmarks and have successfully detected faults in runtime mode. Experimental results have also been produced that confirm the overhead (cost metrics) improvements in our design in compared with the existing works.

VI. References

[1]. R. Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM Journal of Research and Development*, vol. 5, pp. 183-191,1961.

[2]. C. H. Bennett, "Logical Reversibility of Computation," *IBM Journal of Research and Development*, vol. 17, pp. 525–532, 1973.

[3]. R. P. Feynman, "Feynman Lectures on Computation", *Perseus Books*, 1996.

[4]. M. Nielsen and I. Chuang, "Quantum Computation and Quantum Information," *Cambridge University Press*, 2000.

[5]. P. Picton, "Optoelectronic Multi-valued Conservative Logic," *Journal of Optical Computing*, vol. 2, pp. 19-29, 1991.

[6]. R. C. Merkle, "Reversible Electronic Logic using Switches," *Nanotechnology*, vol. 4, pp. 21-40, 1993.

[7]. R. Feynman, "Quantum Mechanical Computers," *Optic News*, vol. 11, pp. 11-20, 1985.

[8]. T. Toffoli, "Reversible Computing," *Technical Memo MIT/LCS/TM-151*, MIT Lab for Computer Science, 1980.

[9]. J. P. Hayes, I. Polian, and B. Becker, "Testing for missing-gate faults in reversible circuits," In *Proc. of Asian Test Symposium*, pp. 100–105, 2004.

[10]. M. Perkowski, J. Biamonte, and M. Lukac, "Test generation and fault localization for quantum circuits," In *Proc. of International Symposium on Multi-Valued Logic*, pp. 62–68, 2005.

[11]. K. Ramasamy, R. Tagare, E. Perkins and M. Perkowski, "Fault localization in reversible circuits is easier than for classical circuits", In *Proc. of International Workshop on Logic and Synthesis*, 2004.

[12]. A. Chakraborty, "Synthesis of reversible circuits for testing with universal test set and C-testability of reversible iterative logic arrays," In *proc. of VLSI Design*, pp. 249-254, 2005.

[13]. H. Rahaman, D. K. Kole, D. K. Das and B. B. Bhattacharya, "Fault Diagnosis for Missing-Gate Fault (SMGF) Model in Reversible Quantum Circuits," *International Journal of Computer and Electrical Engineering (Elsevier)*, vol. 37, pp. 475–485, 2011.

[14]. B. Mondal, D.K. Kole, H. Rahaman and D. K. Das, "Generator for Test Set Construction of SMGF in Reversible Circuit by Boolean difference method," In *proc. of Asian Test Symposium*, pp. 68-73, 2014.

[15]. B. Mondal, C. Bandyopadhyay, D. K. Kole, J. Mathew and H. Rahaman, "Diagnosis of SMGF in ESOP based Reversible Logic Circuit", *International Symposium on Electronic System Design*, pp. 89-93, 2014.

[16]. J. Mondal, D. K. Das, D. K. Kole, H. Rahaman and B.B. Bhattacharya, "On Designing Testable Reversible Circuits Using Gate Duplication", *International Symposium on VLSI Design and Test*, pp. 322-329, 2013.

[17]. D. K. Kole, H. Rahaman, D. K. Das and B. B. Bhattacharya, "Derivation of Optimal Test Set for Detection of Multiple Missing-Gate Faults in Reversible Circuits," In *proc. of Asian Test Symposium*, pp. 33-38, 2010.

[18]. S. N. Mahammad, S. K. S. Hari, S. Shroff and V. Kamakoti, "Constructing Online Testable Circuits Using Reversible Logic," *International Symposium on VLSI Design and Test*, pp. 373-383, 2006.

[19]. D. P. Vasudevan, P. K. Lala, and J. P. Parkerson, "Online testable reversible logic circuit design using NAND blocks," *International Symposium on Defect and Fault-Tolerance in VLSI Systems*, pp. 324–331, 2004.

[20]. D. P. Vasudevan, P. K. Lala, D. Jia, and J. P. Parkerson, "Reversible Logic Design with Online Testability," *IEEE Transactions on Instrumentation and Measurement,* vol. 55, pp. 406–414, 2006.

[21]. S. N. Mahammad, S. Hari, S. Shroff, and V. Kamakoti, "Constructing online testable circuits using reversible logic," *International Symposium on VLSI Design and Test*, pp. 373–383, 2006.

[22]. S. N. Mahammad and K. Veezhinathan, "Constructing online testable circuits using reversible logic," *IEEE Transactions on Instrumentation and Measurement*, vol. 59, pp. 101–109, 2010.

978-1-5090-5163-2/16/$31.00 ©2016 IEEE

[23]. N. Farazmand, M. Zamani, and M. B. Tahoori, "Online Fault Testing of Reversible Logic using Dual Rail Coding," In *proc. of International On-Line Testing Symposium*, pp. 204–205, 2010.

[24]. D. K. Kole, H. Rahaman, and D. K. Das, "Synthesis of Online Testable Reversible Circuit," *International Symposium on Design and Diagnostic of Electronic Circuits and Systems*, pp. 277–280, 2010.

[25]. V. V. Shende, A. K. Prasad, I. L. Markov, and J. P. Hayes, "Synthesis of reversible logic circuits," *IEEE Transaction on Computer Aided Design of Integrated Circuits and Systems*, vol. 22, pp. 723-729, 2003.

[26]. I. Polian, J. P. Hayes, T. Fienn, and B. Becker, "A family of logical fault models for reversible circuits," In *proc. of Asian Test Symposium*, pp. 422-427, 2005.

[27]. M. A. Nashiry, G.G. Bhaskar and J.E. Rice, "Online Testing for Three Fault models in Reversible Circuits," *ISMVL*, 2015.

[28]. R. Wille, D. Grosse, L. Teuber, G. W. Dueck and R. Drechsler, "Revlib: an online resources for reversible functions and reversible circuits," *ISMVL*, vol. 24, pp. 220–225, 2008.

Table 1: Comparison of different online testing approaches

Benchmark name	Approach in [20,19]		Approach in [23]		Approach in [24]		Our approach	
	QC	GO	QC	GO	QC	GO	QC	GO
9symml	25598	5952	32220	532	28644	9	28388	9
apex5	195061	45362	213843	3518	26516	117	20700	117
apla	17815	4142	20574	334	7036	10	6878	10
bw	19019	4422	25140	386	2500	5	1888	5
cm82a	2163	502	2823	50	354	5	310	5
con1	1647	382	1929	38	456	7	414	7
cu	5560	1292	6234	110	2376	14	2298	14
d17	10376	2412	11478	186	3216	10	3120	10
ex2	1733	402	2241	42	312	5	284	5
ex5p	84852	19732	98898	1540	4980	8	3688	8
f51m	128927	29982	164889	2670	76126	14	74802	14
frg2	218109	50722	247332	4048	32384	143	24938	143
max46	15321	3562	19314	326	11102	9	10890	9
misex3	141354	32872	174732	2816	401858	14	398356	14
rd73	16482	3832	20583	342	2446	7	2288	7
sqn	11236	2612	14256	240	4396	7	4246	7
sqrt8	5689	1322	7086	122	1324	8	1246	8
sym9	24652	5732	31890	528	14140	9	13884	9
table3	223054	51872	265566	4294	162102	14	160080	14
z4ml	3883	902	5070	88	1380	6	1286	6
Average	57626.55	13400.5	68304.5	1110.5	39182.4	21.1	37999.2	21.1

Table 2: Improvements achieved by our approach

	Parameters	Existing approaches		
		Approach in [20,19]	Approach in [23]	Approach in [24]
Achieved improvements	QC	34.05%	44.03%	3%
	GO	99.84%	98.10	0%

Table 3: Overhead calculation of the testable design over the non-testable design

Approach	Average overhead for testability feature	
	QC	GO
Approach in [20,19]	312.28%	63409.48%
Approach in [23]	388.67%	5163.03%
Approach in [24]	300.67%	0%
Our approach	100%	0%

Table 4: Comparison Analysis

Benchmark name	GC	Approach in [27]		Our Approach		Our Approach	
		GC	Design Overhead (%)	GC	Design Overhead (%)	Reduced GC	Reduced Design Overhead (%)
apex5	2909	7868	171	5820	102	2048	49
Apla	80	180	125	162	102	18	23
Cm82a	22	54	145	46	102	8	43
Cu	40	108	166	82	102	26	64
Dk17	49	118	137	100	102	18	35
Ex2	15	40	166	32	102	8	64
Frg2	3724	7594	110	7450	102	44	8
Sqrt8	40	96	140	82	102	14	38
Sym9	28	74	164	58	102	16	62
Z4ml	48	110	130	98	102	12	28

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